

❖ **INTRODUCTION:**

Hypernatremia is defined as plasma sodium ≥ 145 mmol/L.

Hypernatremia is not a common electrolyte abnormality in children, but it can be associated with significant neurologic injury especially in those with acute and rapid changes in serum sodium.

Hypernatremia is most often caused by the failure to replace water losses, most commonly due to gastrointestinal fluid loss in children. As a result, water is lost more than sodium, causing increase the plasma sodium concentration

Severity of hypernatremia:

Hypernatremia severity is defined according to the plasma sodium level:

- Mild: Na 146 – 149 mmol/L
- Moderate: Na 150 – 169 mmol/L
- Severe: Na ≥ 170 mmol/L

❖ **ETIOLOGY:**

The following table summarizes common causes of hypernatremia based on the intravascular volume status:

Types of hypernatremia (according to intravascular volume of the child)	Causes
Hypovolemic hypernatremia (most common)	Excess water losses (Vomiting, diarrhea), Renal losses (diabetes mellitus, diuretics), third-space losses
Euvolemic hypernatremia	Central Diabetes insipidus, nephrogenic diabetes insipidus, skin loss (heat stroke, fever, burns), inadequate water intake, hypodipsia
Hypervolemic hypernatremia	Excessive salt intake, hypertonic saline administration, sodium bicarbonate administration, hyperaldosteronism

❖ **EVALUATION:**

Hypernatremia in children should be evaluated to determine the underlying cause. However, in severely ill children, priority is for resuscitation and acute management.

The underlying etiology of hypernatremia is usually evident from the history. Because hypernatremia in children is most often due to unreplaced hypotonic fluid losses, the history

focuses on whether there are increased body fluid losses (e.g., diarrhea) or inadequate fluid intake.

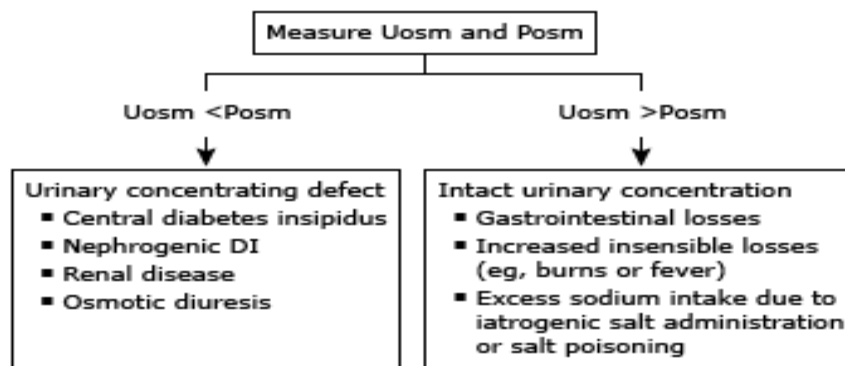
Symptoms and signs:

Irritability, high-pitched cry, lethargy, fever, renal failure, rhabdomyolysis can progress to seizure coma and death in severe cases and can mimic those of infections and sepsis

In neonates: lethargy, irritability reduce urinary frequency, weight loss, fever, doughy skin

Laboratory evaluation:

In cases when the cause of hyponatremia remains uncertain, comparing the urine osmolality (Uosm) with the plasma osmolality (Posm) may be helpful in establishing the underlying etiology.



Blood: serum osmolality, urea and electrolytes, bone profile, and blood gas
(To measure the PH, bicarbonate, and glucose)

Urine: urine osmolality, creatinine and electrolytes in urine, fraction excretion of sodium (FeNa)

Water deprivation test and desmopressin administration DDAVP are used in children with suspected central or nephrogenic diabetes insipidus when the diagnosis is not clear from the available clinical and laboratory data and should only be done after discussion with pediatric endocrinologist and pediatric nephrologist.

Complications of hyponatremia:

Brain hemorrhage, central pontine and extra pontine myelinolysis, thromboembolism, hyperglycemia, and hypocalcemia

❖ **MANAGEMENT:**

The general principle is that changes in plasma sodium that happened acutely can be corrected relatively quickly over 24– 48 hours, whereas chronic changes should be treated slowly.

Where to admit patients with hyponatremia depends on the clinical status of the patient, sodium level, chronicity of hyponatremia and presence of other co-morbidities.

Children with severe hyponatremia serum $\text{Na} \geq 170 \text{ mmol/L}$ should be admitted to the **pediatric intensive care unit (PICU)**, while children with moderate hyponatremia serum $\text{Na } 150\text{-}169 \text{ mmol/L}$ should be admitted to **pediatric high dependency unit HDU**.

Correct hypovolemia with 0.9% NaCl bolus 20 ml/kg over 30 minutes and can be repeated if required (if >40 ml/kg seek senior advice)



- Calculate free water deficit :
 - 1- Total body water (TBW) = % body water (0.7 for children < 2 years of age, 0.6 for children ≥ 2yr) x weight (kg)
 - 2- Free water deficit (FWD) milliliters = TBW x ((current Na – target Na)/target Na*)
- *Target Na = target serum Na level in 24 hours of starting treatment, which should not exceed a fall of sodium greater than 10 mmol/L per day



- Maintenance needs :
- The rule of 100 ml for the first 10 kg, 50 ml for the second 10 kg and 20 ml for the remaining kg per day



- Total fluid requirement in the first 24 hours = FWD + maintenance needs
- *Calculate the rate of total fluid requirement per hour (total /24 hours)



- Calculate the Na maintenance required for the patient.
- *Usually it is 2-3 mmol / kg / day



- Choose IV fluid to replace FWD, maintenance needs and Na requirement
- Choose the type of IV fluid based on the nearest IV fluid solution that contains appropriate Na level that the patient needs per day.
- *Sodium concentration in different IV fluids:

Type of IV fluid	Sodium content
Normal saline (0.9% sodium chloride)	154 mmol / L
0.45 % saline (0.45% sodium chloride)	75 mmol / L
0.19 saline (0.18% sodium chloride)	30 mmol / L

*** Rate of correction:**

- For children with chronic hypernatremia (plasma sodium ≥ 150 mmol /L for greater than 24 hours) or those with acute severe hypernatremia, that a rate of correction should not exceed a fall of sodium greater than 0.5 mmol /L per hour (i.e., 10 to 12 mmol/L per day).

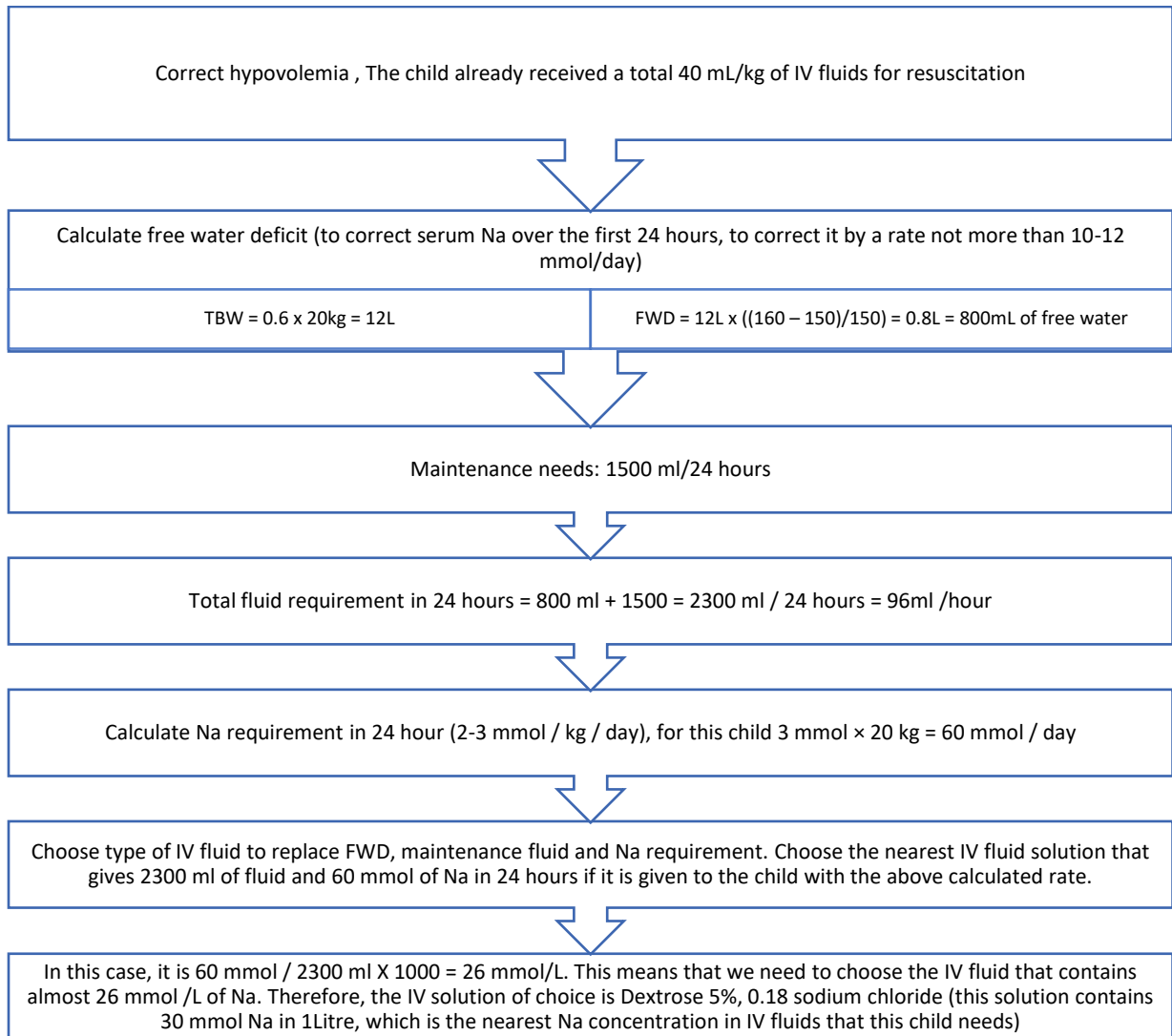
***Ongoing losses:**

- Children may have excess free water losses not accounted for by calculating for maintenance fluids (e.g., continuing diarrhea), and should receive replacement of these ongoing losses to prevent further electrolyte derangement.
- Replace urine, gastrointestinal and other losses 1:1 every 4 hours over next 4 hours (use 0.45% sodium chloride or 0.9% sodium chloride based on the amount of sodium in the lost fluid).

*In cases where hypernatremia alone is the primary abnormality (Euvolemic or hypervolemic hypernatremia) , therapy is aimed at correcting the plasma sodium by providing free water and determining a rate of desired correction.

❖ CLINICAL EXAMPLE:

A 20-kg child is estimated to have a 10 percent dehydration and a serum/plasma sodium concentration of 160 mmol /L. He received a total of 40mL/kg of sodium chloride for resuscitation in the emergency department.



*Replace urine, gastrointestinal and other losses 1:1 every 4 hours over the next 4 hour.

*Repeat serum Na after 2 hours of starting treatment and then every 4 hours.

*Adjust IV fluid type based on the rate of the drop of serum Na. Do not exceed a rate of drop of Na of more than 10-12 mmol/L per 24 hours.

References:

Disorders of Water Metabolism in Children, Michael L. Moritz, Juan Carlos Ayus, Pediatrics In Review (2002), 10.1542/pir.23.11.371

Hyponatremia in children , Michael J Somers, MD, Avram Z Traum, MD , UpToDate , © 2022 UpToDate

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